CHAPTER 2

# Quadratics: forms and choice of method

**Chapter path**

**Objectives.** Choose a form; solve an equation or inequality; read an extremum; prove the formula for the roots.
**Prerequisites.** Special products, zero-product property, sign of a product, square roots.
**Yearly reference point.** 22 September-1 October: 8 hours

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Expand ( $x-3)^2$.
2. Factorise $x^2-16$.
3. Solve ( $x-2)(x+5)=0$.
4. Solve $x^2$ = 9 in $\mathbb{R}$.
5. Give the sign of 2 $x-6$.
6. Determine the minimum of ( $x-1)^2+4$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

**Oral work and logic.** Present two methods for factorising a quadratic. In the proof by discriminant, name the three cases and explain why they are exhaustive.

## Investigate before formalising

A rectangle has an area of 24 m $^2$ and a perimeter of 20 m. Without randomly trying all the dimensions, determine its sides. Start with a drawing and explain the two constraints.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

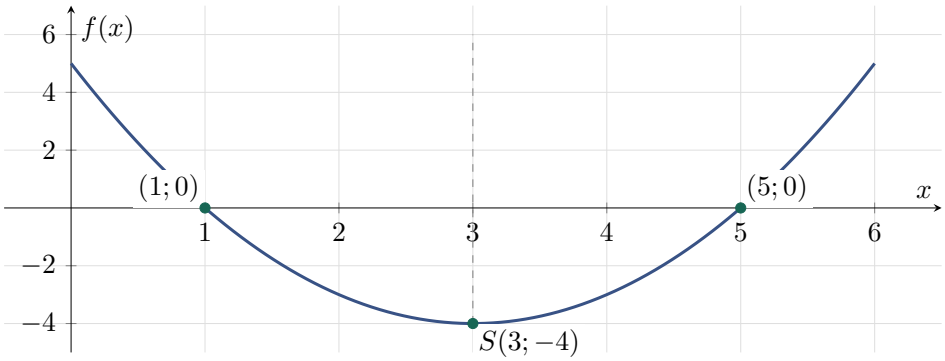
## 2.1 Three forms, three readings

**Definition and results**

A quadratic function can be written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form makes it easy to calculate $f(0)$ ; the vertex form $a(x-\alpha)^2 + \beta$ reveals vertex and extremum; a factorised form $a(x-r_1)(x-r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

**Status and justification.** Completing the square is based on $x^2+2ux = (x+u)^2-u^2$. The existence of the vertex form is proved by expanding; computing it in general is not a routine skill that is required.

**Worked example.** $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is $-4$ at 3 ; the roots are 1 and 5; the y-intercept is 5.




Curve of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the line $x = 3$.

**Watch out**

A factorised expression is not obtained by putting each term separately in brackets.

**Immediate application.** Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and its minimum.

## 2.2 Discriminant and solving

**Definition and results**

For $ax^2+bx+c=0$ with $a\neq 0$, set Delta= $b^2-4ac$. If Delta $<0$, no real root; if Delta= 0, one root $-\frac{b}{2a}$; if Delta $>0$, two roots $\frac{-b-\sqrt{\Delta}}{2a}$ and $\frac{-b+\sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

**Status and justification.** Multiplying the equation by 4 $a\neq 0$, we obtain (2 $ax+b)^2$ = Delta. A square is positive or zero: the three cases follow. If Delta $>0$, $2ax+b=\pm\sqrt{\Delta}$. Splitting into cases avoids taking the square root of a negative number.

**Worked example.** $2x^2+x-3=0$ : $\Delta=1+24=25$ ; $x=\frac{-1\pm 5}{4}$ i.e. 1 and $-\frac{3}{2}$. Check: $(x-1)(2x+3)=2x^2+x-3$.

**Watch out**

The denominator is 2 $a$, not 2 ; the sign of b is taken with its brackets in $b^2$.

**Immediate application.** Solve 3 $x^2-6x+3=0$ and $x^2+2x+5=0$.

## 2.3 Factorising without automatically using the discriminant

**Definition and results**

If two real roots exist, their sum is $-\frac{b}{a}$ and their product $\frac{c}{a}$. An obvious root or a special product often gives a shorter factorisation. Polynomials of degree 2 vanishing at two distinct real numbers r and s are exactly $a(x-r)(x-s)$ with $a\neq 0$.

**Status and justification.** Expanding $a(x-r)(x-s)=ax^2-a(r+s)x+ars$, identifying the coefficients gives the relations. Conversely, if r and s have this sum and this product, expanding gives back the polynomial.

**Worked example.** $x^2-7x+12$ : 3 and 4 have sum 7 and product 12. Thus ( $x-3)(x-4$). To also impose $f(0)=24$, we need 12 $a$ = 24, so $a=2$.

**Watch out**

The sum of the roots is $-\frac{b}{a}$ ; do not forget the minus sign. A single root is not enough to determine a.

**Immediate application.** Determine the quadratic polynomial f such that $f(1)=f(4)=0$ and $f(0)=-8$.

## 2.4 Sign, inequalities and optimisation

**Definition and results**

For two distinct roots $r < s$, the quadratic has the sign of a outside [ $r; s$] and the sign opposite to a between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it has the sign of a away from it. Optimisation is done on the domain of the problem, which may limit access to the vertex.

**Status and justification.** The sign table of the factors $a(x - r)(x - s)$ gives the result. The vertex form compares $f(x)$ à $\beta$ using the sign of $a(x - \alpha)^2$.

**Worked example.** $-2(x - 1)(x - 5) \geq 0$ is equivalent to $x \in [1; 5]$. For $A(x) = x(12 - x)$ on [0; 12], $A(x) = 36 - (x - 6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a maximum area of 36 m $^2$ when it is square.

**Watch out**

In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

**Immediate application.** Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on [0; 3].

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

### Exercise 2.1 · Complete a square

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in the form $(x - 4)^2 + k$, with $k$ a real number to be determined.
2. Deduce a factorisation of $f(x)$ using a special product.
3. Draw the sign table of $f$, then solve in $\mathbb{R}$ the inequality $f(x) \leq 0$.

### Exercise 2.2 · Discussing without heavy calculation

We study the functions $f(x) = x^2 + 4$, $g(x) = (x + 2)^2$ and $h(x) = -3(x - 1)(x + 4)$, all defined on $\mathbb{R}$.

1. For each function, determine any real roots by working directly from the form given.
2. State the sign of each function on $\mathbb{R}$. State explicitly the values at which it vanishes.
3. Explain why computing a discriminant is unnecessary in all three cases.

### Exercise 2.3 · Selecting a Strategy

We want to solve in $\mathbb{R}$ the following three equations:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: special product, finding two numbers with known sum and product, or discriminant. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Check the solutions of the first two equations by substitution. For the third, check the sum and product of the roots obtained.

### Exercise 2.4 · Find a parabola

The curve of a polynomial function $f$ of degree 2 cuts the x-axis at the points with x-coordinates $-2$ and 3. It also passes through the point $A(1; 12)$.

1. Determine an expression for $f(x)$ from these three pieces of information. State the chosen form and why it is suitable.
2. Expand the expression found. Deduce the y-intercept of the curve.
3. Check that the function obtained satisfies the three initial pieces of information.

### Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of fencing to enclose three sides of a rectangular vegetable garden. The fourth side runs along a wall and needs no fencing. Let $x$ the length, in metres, of each of the two sides perpendicular to the wall. The whole length of fencing is used.

1. Draw a diagram and express the length of the side parallel to the wall in terms of $x$. State the values of $x$ for which the rectangle is not degenerate.
2. Express the area $A(x)$, in square metres. Determine its maximum value and the corresponding dimensions, using the method of your choice.
3. Check that these dimensions use exactly 20 m of fencing.

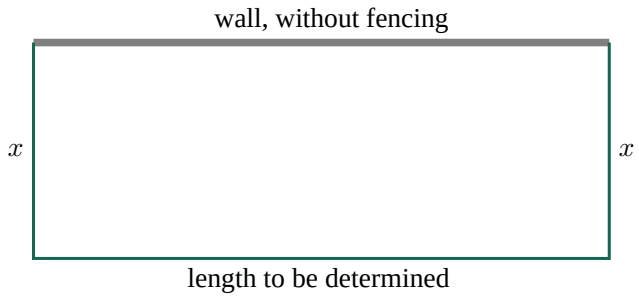



Diagram of the vegetable garden: only the three green lines use the 20 m of fencing. The proportions of the drawing do not determine the optimal dimensions.

### Exercise 2.6 · A Parameter

For every real number $m$, consider the equation with real unknown $x$ :

$$x^2 - 2x + m = 0.$$

The number $m$ is a parameter: it remains fixed when the equation is solved.

1. Determine the values of $m$ for which the equation has two distinct real solutions.
2. Determine the values of $m$ for which it has a unique real solution, then those for which it has none.
3. Give the solution when it is unique. Explain why the three cases considered cover all values of $m$.

### Exercise 2.7 · Link with sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for every natural number $n$. We seek all indices at which the term of the sequence is strictly negative.

1. First solve in $\mathbb{R}$ the inequality $x^2 - 5x + 4 < 0$.
2. Deduce all the natural numbers $n$ such that $u_n < 0$. Justify why a real interval is not the final answer here.

### Exercise 2.8 · Delayed review

A student studies the function $f$ defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She writes: "The minimum is $-5$ and it is attained at $x = 3$. »

1. Say what is correct and what is wrong in this sentence. Rewrite it correctly, justifying your answer.
2. Solve in $\mathbb{R}$ the inequality $f(x) \leq -3$, then check the two bounds of the obtained set.

**Independent work.** Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 43 p. 99, questions a and c.** First find an obvious root, then the other using the product of the roots.

**CQFD, exercise 49 p. 99, questions a and b.** Express the sum and the product as an equation ; do not assume in advance that the numbers exist.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. The roots of $(x+1)(x-4)$ are

   A. 1 and 4

   B. $-1$ and $-4$

   C. $-1$ and 4

   D. 1 and $-4$

2. The discriminant of $x^2 - 2x + 3$ is

   A. $-8$

   B. $8$

   C. $16$

   D. $-2$

3. The vertex of $(x-2)^2 + 5$ is

   A. $(2;-5)$

   B. $(-2;5)$

   C. $(5;2)$

   D. $(2;5)$

4. The sum of the roots of $2x^2 - 6x + 1$ is

   A. $6$

   B. $3$

   C. $-3$

   D. $\dfrac{1}{2}$

5. $x^2 - 4 < 0$ is equivalent to

   A. $x > 2$

   B. $x < -2$

   C. $x \in ]-2;2[$

   D. $x \in [-2;2]$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $x^2 + 6x + 9$ factorises as

   A. $(x-3)^2$

   B. $(x+9)(x+1)$

   C. $(x+6)^2$

   D. $(x+3)^2$

2. The product of the roots of $3x^2 + x - 2$ is

   A. $-\dfrac{2}{3}$

B. $-\dfrac{1}{3}$

C. $\dfrac{2}{3}$

D. $-2$

3. $-x^2 + 4x$ reaches its maximum on [0; 1] at

A. 2

B. 0

C. 1

D. 4

4. A double root requires

A. $a = 0$

B. $\Delta = 0$

C. $b = 0$

D. $c = 0$

5. The quadratic polynomials with roots 0 and 2 are

A. $a(x + 2)$

B. $a(x - 2)^2$

C. $x(x - 2)$ only

D. $ax(x - 2)$, $a \neq 0$

## True or false? Justify.

1. A quadratic function always has two real roots.

2. A quadratic with two distinct roots is negative between them.

3. For every real x, ( $x - 2)^2 \geq 0$.

**Personal review**

I justify the choice of form; I distinguish roots and vertex; I finish an inequality with a set; I respect the domain of the model.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

# Solutions and teaching notes, chapter 2

**Teacher use.** The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $x^2 - 6x + 9$. *Action:* Reconstruct the product of two binomials.
2. $(x - 4)(x + 4)$. *Action:* Use the difference of two squares on two examples.
3. $x$ = 2 or $x = -5$. *Action:* Have students verbalise the zero-product rule and its assumption.
4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the solutions of the equation.
5. Negative before 3, zero at 3, positive after 3. *Action:* Draw the line then make the sign table.
6. 4 attained only at 1. *Action:* Start from the fact that a square is non-negative.

## Investigation

The positive lengths x and y satisfy $x + y$ = 10 and $xy$ = 24. They are the roots of $t^2 - 10t + 24 = (t - 4)(t - 6)$. Dimensions 4 m and 6 m; check the area and the perimeter.

## Direct applications

1. $g(x) = (x + 1)^2 - 4 = (x - 1)(x + 3)$. Roots 1 and $-3$, minimum $-4$ at $-1$.
2. The first is 3($x - 1)^2$ = 0, so 1. For the second, Delta= $-16 < 0$: no real solution.
3. $f(x) = a(x - 1)(x - 4)$ ; $4a = -8$, so $f(x) = -2(x - 1)(x - 4)$.
4. $(x - 5)(x + 1) < 0$ gives ] $-1$; 5[. The vertex of $x(8 - x)$ is at 4, outside the domain; on [0; 3], the function increases and its maximum is 15 at 3.

## Exercises: methods and answers

**Exercise 2.1 - Completing a square.** $x^2 - 8x + 7 = (x - 4)^2 - 9 = (x - 1)(x - 7)$. The leading coefficient is positive; the quadratic is negative or zero between the roots: [1; 7].

**Exercise 2.2 - Discussing without heavy calculation.** $x^2 + 4 > 0$ always, no root. ($x + 2)^2 \geq 0$, double root $-2$. For the last one, roots $-4$ and 1; positive on ] $-$ 4; 1[, negative outside, zero at the bounds.

**Exercise 2.3 - Choose a strategy.** Identity: $x = \pm 3$. Sum and product: 4 and 5. For the third, no obvious integer root: Delta= 17 and $x = \dfrac{3 \pm \sqrt{17}}{4}$. The discriminant applies in all cases, but is not always the most efficient.

**Exercise 2.4 - Finding a parabola.** $f(x) = a(x + 2)(x - 3)$. At 1: $-6a$ = 12, therefore $a = -2$ ; $f(x) = -2x^2 + 2x$ + 12, hence $f(0) = 12$.

**Exercise 2.5 - A vegetable garden against a wall.** The two widths use 2 $x$ metres; length $20 - 2x$. Domain [0; 10] for the function, ]0; 10[ for a non-degenerate rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x-5)^2$. Maximum 50 m $^2$, dimensions 5 m and 10 m.

**Exercise 2.6 - A parameter.** $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$ ; none if $m > 1$. The parameter m stays fixed when solving for x.

**Exercise 2.7 - Link with sequences.** $u_n = (n-1)(n-4)$; negative for 1 $< n <$ 4. Since n is an integer, the only indices are 2 and 3. The set of indices is not the whole real interval.

**Exercise 2.8 - Delayed review.** The minimum is $-5$ at $-3$. The inequality gives ( $x+3)^2 \leq 1$, i.e. $-1 \leq x+3 \leq 1$, so [ $-4; -2]$.

**CQFD, exercise 43 p. 99, questions a and c.** a: obvious root 1, product $-30$, hence the second root $-30$. c: write 4 $x^2 + 9x$ + 5; root $-1$, product $\frac{5}{4}$, hence the second root $-\frac{5}{4}$. Check the sums, respectively $-29$ and $-\frac{9}{4}$.

**CQFD, exercise 49 p. 99, questions a and b.** The numbers are the roots of $t^2 - st + p$. a : $t^2 - t$ + 1, discriminant $-3$: no real pair. b: $t^2 + 60t$ + 756, discriminant 576 = 24 $^2$ : $-18$ and $-42$. Their sum is $-60$ and their product 756.

## MCQ: answers and reasons

**MCQ 1.**

1. C. Set each factor to zero.

2. A. $4 - 12 = -8$.

3. D. The square vanishes at 2, value 5.

4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.

5. C. Positive leading coefficient, strict inequality.

**MCQ 2.**

1. D. Algebraic identity.

2. A. $\frac{c}{a} = -\frac{2}{3}$.

3. C. The function increases up to 2; on this domain, right endpoint 1.

4. B. For a nonzero, a zero discriminant characterises a double root.

5. D. The multiplier is arbitrary and nonzero.

## True or false

1. False: $x^2$ + 1 is strictly positive.

2. False without an assumption on a; $-(x-1)(x-2)$ is positive between 1 and 2.

3. True: the square of a real number is positive or zero; equality only at 2.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.